Every mere proposition is a set

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 321-323 and its proof environment opens on line 324, the next line; the pairing is by adjacency and the retained proof range 324-330 is the whole environment.

Editorial note. The retained counter values (chapter 3, section 3, theorem offset 3) make the counted lemma print as Lemma 3.3.4, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:prop-set = 3.3.4).

Editorial note. One label the proof cites lies outside the card, cor:transport-path-prepost (book Corollary 2.11.2, whose chapter file is not part of the pinned corpus so the pointer span is the label line of this chapter dump), and is declared in external_references as a non-load-bearing navigation pointer: the retained proof states the computed equation it needs.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: every mere proposition is a set (author label thm:prop-set, book Lemma 3.3.4).

Allowed standard prerequisites (editorial).

- Mere proposition. A proof of is-prop(A) assigns to every pair of points a path between them.. It is the hypothesis.
- Dependent action on paths. For a dependent function g the action on paths of g at p is a dependent path between the transports of its values.. It is the step that produces the equation for an arbitrary path.
- Transport in a path family computes by concatenation. Transporting a path in the family of paths out of a point computes as concatenation, so the dependent path becomes an explicit concatenation identity.. It turns the dependent path into the equation that kills both paths.

Source reference. Original corollary 2.11.2: sources/hott-logic/book-labelnumbers-first-edition.tex, source lines 188–188. Book Corollary 2.11.2, the computation of transport in a family of paths by concatenation, invoked at source line 327. Navigation pointer only: the retained proof states the resulting equation. That chapter file is not part of the pinned corpus, so the pointer span is the label line of this chapter dump.

Lemma 3.3.4. *Every mere proposition is a set.*

Proof. Suppose $f : \mathsf{isProp}(A)$; thus for all $x, y : A$ we have $f(x, y) : x = y$. Fix $x : A$ and define $g(y) :\equiv f(x, y)$. Then for any $y, z : A$ and $p : y = z$ we have $\mathsf{apd}_g(p) : p_*(g(y)) = g(z)$. Hence by original corollary 2.11.2, we have $g(y) \cdot p = g(z)$, which is to say that $p = g(y)^{-1} \cdot g(z)$. Thus, for any $p, q : x = y$, we have $p = g(x)^{-1} \cdot g(y) = q$. □